Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Marshall, John Ph.D.		
Project Title:	Microgravity Experiments to Evaluate Electrostatic Forces in Controlling Cohesion and Adhesion of Granular Materials		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Mountain View	**State:**	CA
Zip Code:	94043	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	01/01/1998	**End Date:**	12/31/2004
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Green, Robert D	**Contact Phone:**	216-433-5402
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Flight Program:			
Flight Assignment:	note: end date extended 1 year		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	[Ed. note September 2019--transferred information from 2002 Archives record] Using homogeneous and inhomogeneous electrical-field grain manipulation methods, the flight experiment Electrostatics of Granular Material (EGM) will “interrogate” triboelectrically charged grain populations to determine the magnitude of net charges or monopoles (M) on individual grains, the magnitude of accompanying dipoles (D), and the longevity of the static charges. D/M ratios will thus be determined as a measure of the distribution and relationship of charges on individual grains and populations of grains. This knowledge will add a dimension to fundamental electrostatic theory of particulates, since fixed dipoles are not widely treated in this field. Flight data, along with computational codes, will be applied to understanding the role of interparticle forces in mediating the behavior of particulate clouds in nature and industrial situations (e.g., via aggregation versus dispersion mechanisms). Also, the data will be applied to the adhesion of grains to surfaces, and the cohesion or internal Coulombic friction of granular masses.		

	At the end of 2001, the project transitioned into a ground-based effort as a result of recent changes in the Microgravity Program. The current effort is proceeding under the assumption that EGM may fly at a later date. The research has two themes; the first is to gain a better understanding of electrostatic phenomena as an aid to ultimately improving the design of the flight experiment, as well as better defining the science to be studied in microgravity. The second theme is the acquisition of knowledge about processes that cannot be studied in EGM, but which nevertheless provide scientific context to the EGM results. For example, studies (both experimental and modeling) are being conducted on the causes of triboelectrification, the causes of charge biasing in grain masses, the initial fast decay of some charged systems, and the end-member cases of pure dipole and pure monopole interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)